

# Artem Semidetnov

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## Education

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| <b>MSc</b> | <b>Geneva University</b> , Mathematics                                           | 2025 – 2027 |
| <b>BSc</b> | <b>Saint-Petersburg State University</b> , Mathematics                           | 2021 – 2025 |
| <b>BSc</b> | <b>Neapolis University Of Pafos</b> , Applied Computer Science JetBrains Program | 2022 – 2024 |

## Technical Skills

- **Formal verification:** Arend, dependent type theory, theorem formalization, typechecking, and IDE integration.
- **Programming languages:** Kotlin, Python, SQL, Java, Lean, Arend, Haskell.
- **AI and machine learning:** Coding-agent frameworks (LangGraph, Koog), model fine-tuning, code completion, and MCP protocol.
- **Spoken languages:** English C1 (IELTS 8.0/9).

## Work Experience

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| <b>JetBrains</b> , Research Intern in HoTT and Dependent Types Lab                                                                                                                                                                                                                                                                                                                                                                           | Remote                |
| <ul style="list-style-type: none"> <li>• Built AI-assisted development features for Arend, including MCP servers exposing typechecking and IDE functionality; contributed to the Arend plugin for IntelliJ IDEA.</li> <li>• Built an experimental LLM coding agent for generating Arend code using the Koog framework.</li> <li>• Fine-tuned models for Arend code completion.</li> </ul>                                                    | Nov 2025 – Apr 2026   |
| <b>JetBrains</b> , Research Intern in HoTT and Dependent Types Lab                                                                                                                                                                                                                                                                                                                                                                           | Pafos, Cyprus         |
| <ul style="list-style-type: none"> <li>• Contributed formalized results in algebra and homotopy type theory to the official Arend library.</li> <li>• Formalized the Eckmann-Hilton argument, Eilenberg-MacLane spaces, and homology of types.</li> <li>• Formalized group automorphisms, Schur's Lemma, Maschke's Lemma, and characterizations of group actions; selected contributions shipped in <a href="#">Arend 1.10</a> ↗.</li> </ul> | June 2024 – Sept 2024 |
| <b>IPONWEB (acquired by Criteo)</b> , Machine Learning Intern                                                                                                                                                                                                                                                                                                                                                                                | Pafos, Cyprus         |
| <ul style="list-style-type: none"> <li>• Reverse-engineered the behavior of a Criteo ML system that analyzes websites and produces word2vec representations; evaluated potential applications.</li> </ul>                                                                                                                                                                                                                                    | May 2023 – Sept 2023  |

## AI-Assisted Mathematical Research

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| <b>On Some More Problems from the Kourouka Notebook</b> ↗                                                                                    | Aug 2026 |
| Vasily Ionin, <b>Artem Semidetnov</b><br><a href="https://doi.org/10.48550/arXiv.2608.29219">https://doi.org/10.48550/arXiv.2608.29219</a> ↗ |          |
| <b>The stable commutator length of a relator is not a one-relator group invariant</b> ↗                                                      | Aug 2026 |
| <b>Artem Semidetnov</b><br><a href="https://doi.org/10.48550/arXiv.2608.21465">https://doi.org/10.48550/arXiv.2608.21465</a> ↗               |          |

## Publications

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| <b>The operad associated to a crossed simplicial group</b> ↗                                                                   | Dec 2025 |
| <b>Artem Semidetnov</b><br><a href="https://doi.org/10.48550/arXiv.2512.14498">https://doi.org/10.48550/arXiv.2512.14498</a> ↗ |          |
| <b>On the geometry of free nilpotent groups</b> ↗                                                                              | May 2021 |
| <b>Artem Semidetnov</b> , Ruslan Magdiev                                                                                       |          |

<https://doi.org/10.48550/arXiv.2106.00095> 